Aaron Low Weng Soon

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SUMMARY

Senior robotics engineer with six years in autonomous vehicles, specialising in multi-sensor calibration and fusion. During my career, I've worked on systems that utilise a combination of deep learning and classical methods like factor graphs, multi-view geometry, and SLAM. I work end to end, developing the algorithms and building the scalable, automated tooling and infrastructure that makes them fast to iterate on and reliable to run.

WORK EXPERIENCE

Wayve — 2024 - Present

Senior Robotics Engineer (Sensor Calibration and Fusion) • *2024* — **London, United Kingdom**

• Designed and implemented multi-sensor calibration algorithms (camera, IMU, lidar) combining deep learning with multi-view geometry and SLAM.

• Built scalable automated pipelines running calibration continuously supporting a fleet of vehicles spread across multiple locations globally.

  • Our improvements led to a 4x speedup in turnaround time for calibrating a vehicle.

• Built an end-to-end deep learning pipeline for training and evaluating visual feature extraction and calibration models, covering dataset generation, curation, and loading.

• Implemented metrics and automated validation that detect miscalibration and drift, catching issues before they reached downstream users.

• Developed internal tooling to deploy, track, and maintain per-vehicle calibrations across the fleet.

• Led technical direction, mentored junior engineers, and helped shape hiring decisions

Motional — 2020 - 2024

Senior Research Engineer (Sensor Calibration) • *2023 - 2024* — **Singapore**

Research Engineer (Sensor Calibration) • *2020 - 2023*

• Developed solutions for sensor (cameras, LIDARs, radars, IMU) calibration and validation of autonomous vehicles.

• Researched deep learning solutions for sensor calibration as well as implementing the infrastructure for training and evaluation.

• Built and maintained cloud-based systems for continuous validation of vehicle sensor data.

  • Our automated system ran daily on a fleet of hundreds of vehicles allowing data-driven calibrations.

• Automated dataset generation and curation for algorithm research.

• Worked on vehicle hardware to deploy online sensor calibration algorithms.

Hyundai-Aptiv Autonomous Driving Joint Venture — 2020

Autonomous Vehicle Intern (Sensor Calibration) — **Singapore**

• Researched deep learning for camera-to-LiDAR calibration and deployed a **RegNet** based model into production with TensorRT.

PUBLICATIONS

• **UniCal: a Single-Branch Transformer-Based Model for Camera-to-LiDAR Calibration and Validation** — a Transformer architecture using self-attention to perform camera-to-LiDAR calibration and validation in a single branch.

• **Depth to Colour Translation for 3D Hand Pose Estimation From Monocular RGB With Generative Adversarial Networks** — MEng thesis, Imperial College London.

EDUCATION

Imperial College London **2018 - 2019**

PhD Researcher, Machine Learning and Computer Vision

• **Research area:** 3D Pose Estimation

Imperial College London **2014 - 2018**

Electrical and Electronic Engineering MEng

• **First Class Honours** • **Dean's List (top 10% of class) Year 2**

TEACHING

HELP University **2019 - 2020**

Lecturer, Faculty of Computing and Digital Technology

• **Lectured introductory programming** and **presented deep learning tutorials to both staff and students.**

PROJECTS

Autonomous Snack Delivery Android (ASDA) **2017**

Robot Navigation

• Autonomous robot that manoeuvres a building and takes an elevator to obtain and deliver snacks.

ParkWare **2016**

Machine Learning and Web

• **Prize Winner ("Best use of Amazon Web Services")** Parking space detection web service using machine learning to detect cars in parking lots.

Without Abandonware **2021**

Unity/C# Developer

• **Featured by GitHub** in "Top entries from Game Off 2021". A 2D platformer written in Unity with C#.

TECHNICAL SKILLS

Programming: Python • C++ • C#

ML and Robotics: PyTorch • TensorFlow • ROS • TensorRT • GTSAM

Infrastructure: AWS • Azure • Kubernetes • Spark • Terraform • Docker • Jenkins • Serverless • Bazel • CMake

Tools and Data: Git • Linux • Bash • Pandas • SQL • Unity

Languages: English (Native) • Malay (Limited Working)